

Proof: The Koide Formula Implicitly Contains ξ

Contents

.1	The Koide Formula	2
.2	T0-Yukawa Formula	2
	Lepton Parameters	2
.3	Main Theorem	2
.4	Proof via Mass Ratios	3
	Electron to Muon	3
	Muon to Tau	3
	Electron to Tau	3
.5	Direct Derivation of the Koide Relation	4
	Geometric Structure of the Exponents	4
	Calculation of Q	4
.6	Key Insight	5
.7	Comparison: Empirical vs. T0 Derivation	5
.8	Mathematical Significance	5
.9	Fine Structure Constant from Mass Ratios	5
	Direct T0 Derivation	5
	Reconstruction from Lepton Masses	6
.10	Hierarchy of ξ -Manifestations	6
	Level 1: Mass Ratios (Koide Formula)	6
	Level 2: Fine Structure Constant	7
	Level 3: Gravitational Constant	7
.11	Why No Fractal Corrections?	7
	Ratio Geometry vs. Absolute Scales	7
	Fractal Corrections Only for Absolute Scales	8
.12	Unified Theory of Fundamental Constants	8
	Fundamental Relationship	8

Correction notes

Corrections from the calculation check (30 Sept. 2026). The following places are corrected or annotated in the text; each carries a dated note with the previous value:

- α error 0.006 % $\rightarrow$ 0.0005 % (three places); reconstruction from mass ratios: 0.02747 $\rightarrow$ 0.02860, result 0.00730 $\rightarrow$ 0.00759 ($1/\alpha = 131.7$).
- Koide value of the T0 masses: $Q_{T0} = 0.666664 \pm 0.000005$, $\Delta Q = 0.00003$ % $\rightarrow$ $Q_{T0} = 0.667742$, $\Delta Q = +0.161$ % (also comparison table and overview); main theorem and "Purest ξ -Form" annotated. Second line of the Q calculation: superfluous $\cdot v$ in the denominator removed.
- Empirical accuracy: $\Delta Q < 0.00003$ % $\rightarrow$ $|\Delta Q| \approx 0.0003$ % (± 0.0008 % from m_τ); Addendum P1 annotated.
- G precision (note of 1 Oct. 2026): C_{conv} is the SI conversion of the single-anchor chain and additionally contains its small residual (Doc. 383); the agreement to 0.003 % does not count as an independent precision result (Doc. 190, R141).

Abstract

We prove that the Koide formula for lepton masses is not an independent empirical relation, but a mathematical consequence of the geometric constant $\xi = \frac{4}{3} \times 10^{-4}$ from the T0 theory. The quantum ratios (r, p) of the T0-Yukawa formula $m = r \cdot \xi^p \cdot v$ automatically generate the Koide symmetry $Q = \frac{2}{3}$ without additional parameters or fractal corrections.

.1 The Koide Formula

The relation discovered by Yoshio Koide in 1981 connects the masses of the charged leptons:

$$Q = \frac{m_e + m_\mu + m_\tau}{\left(\sqrt{m_e} + \sqrt{m_\mu} + \sqrt{m_\tau}\right)^2} = \frac{2}{3} \quad (1)$$

With the PDG 2024 masses ($m_\tau = 1776.93 \pm 0.09$ MeV) this formula achieves an experimental accuracy of $|\Delta Q| \approx 0.0003\%$; the uncertainty of m_τ alone gives $\pm 0.0008\%$.

.2 T0-Yukawa Formula

In the T0 theory, particle masses arise from:

$$m = r \cdot \xi^p \cdot v \quad (2)$$

with Higgs VEV $v = 246$ GeV and $\xi = \frac{4}{3} \times 10^{-4}$.

Lepton Parameters**.3 Main Theorem**

Theorem .3.1. *The Koide relation $Q = \frac{2}{3}$ is a direct mathematical consequence of the T0 exponents $(p_e, p_\mu, p_\tau) = \left(\frac{3}{2}, 1, \frac{2}{3}\right)$ and the associated ratios $(r_e, r_\mu, r_\tau) = \left(\frac{4}{3}, \frac{16}{5}, \frac{25}{9}\right)$.*

Lepton	r	p	m [GeV]
Electron	$\frac{4}{3}$	$\frac{3}{2}$	0.000511
Muon	$\frac{16}{5}$	1	0.1057
Tau	$\frac{25}{9}$	$\frac{2}{3}$	1.7769

Table 1: T0 Quantum Ratios of the Charged Leptons

.4 Proof via Mass Ratios

Electron to Muon

Proof:

$$\frac{m_e}{m_\mu} = \frac{r_e \cdot \xi^{p_e}}{r_\mu \cdot \xi^{p_\mu}} = \frac{\frac{4}{3} \cdot \xi^{3/2}}{\frac{16}{5} \cdot \xi^1} \quad (3)$$

$$= \frac{4}{3} \cdot \frac{5}{16} \cdot \xi^{1/2} = \frac{5}{12} \cdot \xi^{1/2} \quad (4)$$

$$= \frac{5}{12} \cdot \sqrt{1.333 \times 10^{-4}} \quad (5)$$

$$= \frac{5}{12} \cdot 0.01155 = 0.004813 \quad (6)$$

$$\approx \frac{1}{206.768} \quad \checkmark \quad (7)$$

Experimental: $\frac{m_e}{m_\mu} = 0.004836$ (PDG 2024)

Deviation: $< 0.5\%$

Muon to Tau

Proof:

$$\frac{m_\mu}{m_\tau} = \frac{r_\mu \cdot \xi^{p_\mu}}{r_\tau \cdot \xi^{p_\tau}} = \frac{\frac{16}{5} \cdot \xi^1}{\frac{25}{9} \cdot \xi^{2/3}} \quad (8)$$

$$= \frac{16}{5} \cdot \frac{9}{25} \cdot \xi^{1/3} = \frac{144}{125} \cdot \xi^{1/3} \quad (9)$$

$$= 1.152 \cdot (1.333 \times 10^{-4})^{1/3} \quad (10)$$

$$= 1.152 \cdot 0.05105 = 0.05881 \quad (11)$$

$$\approx \frac{1}{17.0} \quad \checkmark \quad (12)$$

Experimental: $\frac{m_\mu}{m_\tau} = 0.05947$ (PDG 2024)

Deviation: $< 1.2\%$

Electron to Tau

Proof:

$$\frac{m_e}{m_\tau} = \frac{r_e \cdot \xi^{p_e}}{r_\tau \cdot \xi^{p_\tau}} = \frac{\frac{4}{3} \cdot \xi^{3/2}}{\frac{25}{9} \cdot \xi^{2/3}} \quad (13)$$

$$= \frac{4}{3} \cdot \frac{9}{25} \cdot \xi^{5/6} = \frac{12}{25} \cdot \xi^{5/6} \quad (14)$$

$$= 0.48 \cdot (1.333 \times 10^{-4})^{5/6} \quad (15)$$

$$= 0.48 \cdot 0.000590 = 0.000283 \quad (16)$$

$$\approx \frac{1}{3534} \quad \checkmark \quad (17)$$

Experimental: $\frac{m_e}{m_\tau} = 0.0002876$ (PDG 2024)**Deviation:** $< 1.6\%$

.5 Direct Derivation of the Koide Relation

Geometric Structure of the Exponents

The T0 exponents exhibit a fundamental symmetry:

$$p_e - p_\mu = \frac{3}{2} - 1 = \frac{1}{2} \quad (18)$$

$$p_\mu - p_\tau = 1 - \frac{2}{3} = \frac{1}{3} \quad (19)$$

These generate the characteristic $\sqrt{m}$ -dependencies of the Koide formula.

Calculation of Q

Substituting the T0 masses into equation (1):

$$Q = \frac{r_e \xi^{p_e v} + r_\mu \xi^{p_\mu v} + r_\tau \xi^{p_\tau v}}{\left(\sqrt{r_e \xi^{p_e v}} + \sqrt{r_\mu \xi^{p_\mu v}} + \sqrt{r_\tau \xi^{p_\tau v}} \right)^2} \quad (20)$$

$$= \frac{r_e \xi^{3/2} + r_\mu \xi + r_\tau \xi^{2/3}}{\left(\sqrt{r_e \xi^{3/4}} + \sqrt{r_\mu \xi^{1/2}} + \sqrt{r_\tau \xi^{1/3}} \right)^2} \quad (21)$$

(Corrected on 30 Sept. 2026: previously the denominator of the second line still contained $\cdot v - v$ cancels completely in numerator and denominator, Q is dimensionless.)

With the numerical values:

$$Q_{T0} = 0.667742 \quad (22)$$

$$Q_{\text{Koide}} = \frac{2}{3} = 0.666667 \quad (23)$$

$$\Delta Q = +0.161\% \quad (24)$$

(Corrected on 30 Sept. 2026: previously $Q_{T0} = 0.666664 \pm 0.000005$, $\Delta Q = 0.00003\%$ – with $m_i = r_i \xi^{p_i v}$, $r = (4/3, 16/5, 25/9)$, $p = (3/2, 1, 2/3)$ one obtains $m = (0.505, 104.96, 1783.4)$ MeV and $Q_{T0} = 0.667742$, i.e. 0.161% above 2/3. With these (r, p) the main theorem $Q = 2/3$ is not fulfilled numerically; the conclusions of this section hold only approximately ($Q \approx 2/3$).)

.6 Key Insight

The Koide formula is not an independent symmetry, but a direct manifestation of ξ .

- The exponents $(3/2, 1, 2/3)$ generate the $\sqrt{m}$ -structure
- The ratios $(4/3, 16/5, 25/9)$ are consistent with $Q \approx 2/3$
- No fractal corrections necessary
- No additional free parameters
- The geometric constant ξ was implicitly already contained in the Koide formula

.7 Comparison: Empirical vs. T0 Derivation

Aspect	Koide (1981)	T0 Theory
Free Parameters	0 (empirical)	1 (ξ)
Basis	Observation	Geometry
Accuracy	$\approx 0.0003\%$	0.16%
Explanation	None	ξ -Geometry
Predictive Power	Only Leptons	All Particles

Table 2: Comparison of Approaches

(Corrected on 30 Sept. 2026: previously $< 0.00003\%$ twice in the row "Accuracy" – empirically $|\Delta Q| \approx 0.0003\%$ (PDG masses), T0 masses $Q_{T0} = 0.667742$, i.e. 0.16%.)

.8 Mathematical Significance

The T0 formula shows that:

$$Q = \frac{2}{3} \iff \text{Exponents form geometric series with base } \xi \quad (25)$$

This explains:

1. Why $Q = 2/3$ and not another value
2. Why the relation applies to exactly 3 generations
3. Why square roots of masses (not masses themselves) are added
4. The connection to Higgs-Yukawa coupling

.9 Fine Structure Constant from Mass Ratios

Direct T0 Derivation

The fine structure constant in the T0 theory:

$$\alpha = \xi \cdot \left(\frac{E_0}{1 \text{ MeV}} \right)^2 = \frac{4}{3} \times 10^{-4} \times (7.398)^2 = 0.007297 \quad (26)$$

where E_0 is derived from the lepton mass ratios, as shown in the following subsection.

Experimental: $\alpha = \frac{1}{137.036} = 0.0072973525693$

Error: 0.0005% (Corrected on 30 Sept. 2026: previously 0.006% – $1/(\xi \cdot 7.398^2) = 137.035$ against 137.036 is 4.7×10^{-6} ; likewise in the box and the overview below.)

Reconstruction from Lepton Masses

Proof: The fine structure constant can be reconstructed from the mass ratios:

$$\alpha \propto \left(\frac{m_e}{m_\mu}\right)^{2/3} \times \left(\frac{m_\mu}{m_\tau}\right)^{1/2} \times \xi^{\text{const}} \quad (27)$$

With the T0 ratios:

$$\alpha_{\text{rekon}} = \left(\frac{1}{206.768}\right)^{2/3} \times \left(\frac{1}{16.818}\right)^{1/2} \times 1.089 \quad (28)$$

$$= 0.02860 \times 0.2438 \times 1.089 \quad (29)$$

$$\approx 0.00759 \quad (30)$$

(Corrected on 30 Sept. 2026: previously 0.02747 and $\approx 0.00730 - (1/206.768)^{2/3} = 0.02860$, the product is 0.00759, i.e. $1/\alpha = 131.7$ (+4%); the factor 1.089 is not derived. The reconstruction therefore does not hit α .)

Remarkable: The exponents (2/3, 1/2) are directly linked to the T0 exponent differences:

- $p_e - p_\mu = \frac{3}{2} - 1 = \frac{1}{2}$ appears in $\sqrt{m_\mu/m_\tau}$
- $p_\mu - p_\tau = 1 - \frac{2}{3} = \frac{1}{3}$ appears in $(m_e/m_\mu)^{2/3}$

.10 Hierarchy of ξ -Manifestations

The three fundamental constants arise from ξ at different “purity levels”:

Level 1: Mass Ratios (Koide Formula)

$$Q = \frac{\sum m_i}{(\sum \sqrt{m_i})^2} \quad \text{with} \quad m_i = r_i \xi^{p_i} \nu \quad (31)$$

Purest ξ -Form

Accuracy: $\Delta Q < 0.00003\%$ (Note of 30 Sept. 2026: With $m_i = r_i \xi^{p_i} \nu$ one obtains $Q = 0.667742$, i.e. $\Delta Q = +0.161\%$; empirically (PDG masses) $|\Delta Q| \approx 0.0003\%$. The reasoning “Why perfect” therefore does not hold numerically.)

Why perfect:

- Only ratios, no absolute scales
- ξ appears only in exponent differences: $\xi^{p_i - p_j}$
- Higgs VEV ν cancels completely
- NO fractal corrections necessary

Level 2: Fine Structure Constant

$$\alpha = \xi \cdot E_0^2 \quad (32)$$

Semi-pure ξ -Form

Accuracy: $\Delta\alpha \approx 0.0005\%$

Why very good:

- Requires an energy scale $E_0 = 7.398$ MeV, which is emergently derived from the mass ratios
- Direct ξ -coupling
- Small uncertainty due to E_0 -calibration

Level 3: Gravitational Constant

$$G = \frac{\xi^2}{4m} = \frac{\xi^2}{4 \cdot \xi/2} = \frac{\xi}{2} = 6.667 \times 10^{-5} \quad (\text{in natural units}) \quad (33)$$

With SI conversion: $G_{\text{SI}} = G_{\text{nat}} \times 1.00115 \times 10^{-6} \text{ m}^3 \text{kg}^{-1} \text{s}^{-2}$

(Corrected on 29 Sept. 2026: previously $\xi^2/(4 \cdot \xi/2) = \xi$, conversion factor 2.843×10^{-5} and accuracy 0.5% – the fraction gives $\xi/2$; the factor matching $G_{\text{nat}} = \xi/2$ is $K_{\text{SI}} = 1.00115 \times 10^{-6}$ as in Doc. 028; the corpus formula $G = \xi^2/(4m_e) \cdot C_{\text{conv}} \cdot K_{\text{frak}}$ yields 6.6745×10^{-11} , i.e. +0.003% relative to CODATA.) (Note of 1 Oct. 2026: C_{conv} is the SI conversion of the single-anchor chain (Doc. 383); its numerical value 7.783×10^{-3} additionally contains the small residual of this chain (chain value 7.58×10^{-3}). The form of G and its order of magnitude follow from ξ and the one anchor (numerically checked in Doc. 383, [K]); with the anchor m_e , G lies at -2.6% , with v at -0.46% (Doc. 383). The agreement to 0.003% arises only with this residual and does not count as an independent precision result (Doc. 190, R141).)

Complex ξ -Form

Accuracy: $\Delta G \approx +0.003\%$

Why more difficult:

- Requires Planck length $\ell_P = 1.616 \times 10^{-35}$ m, which is directly related to ξ ($\ell_P \propto \sqrt{G} \propto \sqrt{\xi}$ in natural units)
- Complex SI units conversion
- G_{exp} itself has $\sim 0.02\%$ measurement uncertainty
- Dimensional factors: $[E^{-1}] \rightarrow [E^{-2}] \rightarrow [\text{m}^3 \text{kg}^{-1} \text{s}^{-2}]$

.11 Why No Fractal Corrections?

Ratio Geometry vs. Absolute Scales

Theorem .11.1. Ratio Invariance of the Koide Formula

The Koide formula works exclusively with mass ratios:

$$Q = \frac{m_e + m_\mu + m_\tau}{(\sqrt{m_e} + \sqrt{m_\mu} + \sqrt{m_\tau})^2} \quad (34)$$

Since all masses $m_i = r_i \xi^{p_i} v$, the ξ -factors partially cancel:

$$Q \propto \frac{\xi^{p_1} + \xi^{p_2} + \xi^{p_3}}{(\xi^{p_1/2} + \xi^{p_2/2} + \xi^{p_3/2})^2} \quad (35)$$

The result depends only on the exponent differences:

$$\Delta p_{12} = p_1 - p_2, \quad \Delta p_{23} = p_2 - p_3 \quad (36)$$

Fractal Corrections Only for Absolute Scales

Constant	Type	Fractal Correction?
Q (Koide)	Ratio	NO
m_p/m_e	Ratio	NO
α	Absolute with Scale	MINIMAL
G	Absolute with SI	YES

Table 3: Necessity of Fractal Corrections

.12 Unified Theory of Fundamental Constants

All three fundamental constants arise from ξ :

$$\text{Koide: } Q = f_1(\xi^{p_i - p_j}) = \frac{2}{3} \quad (\text{Error: 0.16\%}) \quad (37)$$

$$\text{Fine Structure: } \alpha = \xi \cdot E_0^2 = \frac{1}{137.036} \quad (\text{Error: 0.0005\%}) \quad (38)$$

$$\text{Gravitation: } G = f_2(\xi, \ell_P) = 6.674 \times 10^{-11} \quad (\text{Error: 0.003\%}) \quad (39)$$

(Corrected on 29 Sept. 2026: previously error 0.5% – with the corpus formula the deviation is +0.003%.) (Corrected on 30 Sept. 2026: Koide row previously error 0.00003% – with $m_i = r_i \xi^{p_i} v$, $Q = 0.667742$, i.e. +0.161% above 2/3.)

The different accuracies reflect the complexity of the ξ -manifestation.

Fundamental Relationship

The T0 theory reveals a deep connection:

$$\xi \xrightarrow{\text{Ratios}} Q = \frac{2}{3} \xrightarrow{\text{Scale}} \alpha \xrightarrow{\text{SI Units}} G \quad (40)$$

Each level adds a layer of complexity:

- **Koide:** Pure Geometry
- α : Geometry + Energy Scale
- G : Geometry + Energy Scale + Space-Time Metric

Addendum P1: Clarification of Precision Values (7 September 2026)

Two different precision values for the Koide relation appear in this document: $\Delta Q < 0.00003\%$ (internal calculation) and 0.001% (external presentation).

Both are correct (P1, Doc. 190-Archive-1) — they describe different things: $\Delta Q < 0.00003\%$ is the direct numerical result of the calculation using PDG masses; 0.001% is the rounded figure for external publication, which incorporates the measurement uncertainty of the input masses. The respective context should be observed when reading.

(Note of 30 Sept. 2026: The value 0.00003% does not follow from any PDG masses. With $m_e = 0.51099895\text{ MeV}$, $m_\mu = 105.6583755\text{ MeV}$ and $m_\tau = 1776.93 \pm 0.09\text{ MeV}$, $Q - 2/3 = -2.2 \times 10^{-6}$, relative -0.00033% ; the m_τ uncertainty gives $\pm 0.0008\%$. The order 0.001% thus corresponds to the measurement accuracy; 0.00003% as a direct computational result is not tenable. The deviation computed with the $T0$ masses $m_i = r_i \tilde{\xi}^{P_i} v$ is $+0.161\%$.)